



Revetments of LaRabida and Promontory Point

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1 Introduction

Coastal environments are generally the most common victims of erosion, primarily from the continuous onslaught of winds and waves on the area. Major cities found along shorelines have their hands full with protecting the land on which they are situated, and solutions need to be produced. Revetments and breakwater are one of the most common means to protect land from erosion, and disperse energy coming from water based sources efficiently. A revetment is a manmade structure backed by land which is used to combat coastal erosion, while a breakwater is similar in purpose but situated further into the body of water which it is located. Contentious discourse on the best type of revetment to implement in a community is difficult to constrain to a definite answer, primarily as there are a plethora of factors to consider when making the decision. Some of the larger ones include cost effectiveness, material, shape and volume, a metric of erosion prevention effectiveness, and ability to foster and promote local ecosystems. Making basic assumptions as insight from the first few factors, this project aimed to study the implications these revetments have on local biodiversity, and if we can identify specific traits that make one revetment the most effective to implement to promote biodiversity in addition to the other important factors mentioned above.

1.1 History

As a neighborhood situated on Lake Michigan, Hyde Park is guarded on the east by a pretty heavily enforced revetment system. Throughout Chicago's history, these revetments have stood to protect the area from flooding and erosion due to the lake's natural tendency to form large waves moving westwards. The Promontory Point (Fig. 1a) is located at the eastmost end of 55th St., and is

a public park for recreational use. Originally built as a landfill in 1909 by product of Daniel Burnham, there was an implementation of large limestone revetments during the 1920s and 30s, which followed with a transition to a public place of recreation in 1937 by Alfred Caldwell



(a) Promontory Point



(b) LaRabida

Figure 1: Areas of Study

The LaRabida Children's Hospital (Fig. 1b) is situated in Jackson Park just south of 63rd St. Beach, and has a less known history. Located on Silurian age dolostone, which consists of calcified tropical reefs from 400 mya, the Ohmsted Monastery was built where LaRabida currently resides, and was an exhibit of the 1893 Columbian Exposition in Chicago. The Monastery was turned into a children's sanatorium, which burned down in 1922, and was rebuilt in 1932 as the LaRabida Children's Hospital. It was difficult to constrain the timeline of revetment implementation here, but following the Silurian dolostone, there is a rocky pebble beach, as well as the concrete blocks under the Ohmsted Wall. The two center revetments have no information available about when these were implemented, but are assumed to be most recent.

2 Materials and Methods

2.1 Sample Collection

To collect our data, we focused our efforts on field work at two sites, The Promontory Point (PP), and LaRabida Children's Hospital (LR). Between these two sites, we had divided the space covered into six sections, being LR1 through LR5, and PP. Materials of each site were noted based off of one of the group members experience with geology for generalizations, as explicit info was not given for each site. Field work was conducted twice a week for four weeks, on Mondays and Wednesdays

from 14:00-15:30 excluding transit times. The first three weeks involved visiting either LR or PP and collecting all uniquely distinguishable specimens. Collection included noting site location, morphological description, a picture of the specimen, and uncontaminated collection in a vial or bag.

2.2 Species Identification

Species identification relied on DNA and morphological analysis. DNA analysis was completed by extracting DNA from tissue of our samples using Qiagen DNeasy extraction kits, following protocols from the manufacturer and protocols from course materials (QIAGEN, 2006). PCR amplification was performed according to protocols from course materials, targeting the COI region for animals and the rbCL region for plants. Amplified products were submitted for sequencing to the University of Chicago core facility. After sequencing was complete, DNA sequences were compared with known literature sequences using the DNA subway BLAST feature. Dichotomous keys were used for morphological analysis. The Field Museum's guide for Chicago mosses was especially useful for moss identification (Strauss et al., 2022). In addition to these guides, iNaturalist was also used for species identification (iNaturalist, 2025). Crowd-sourced and algorithmic identification helped to suggest new species identifications, and comparison with iNaturalist's vast bank of photographic observation confirmed hypothesized identifications. Species were given native or exotic status based on DNA subway and using data from the USDA PLANTS database and Kew Plants of the World online, and information on invasiveness was determined by these databases and the Midwest Invasive Species Information Network (Kew Royal Botanic Gardens, 2025; Michigan State University, 2025; U.S. Department of Agriculture, Natural Resources Conservation Service, 2025). Phylogenies were constructed using DNA subway.

2.3 Organism Surveying

The final week of data collection focused on morphological analysis through species richness to further determine biodiversity metrics of each revetment. Using an ArcGIS Survey 123 form, we noted every sample found along LR1-LR5 at transects of 0, 10, and 30 feet from the shoreline. Each entry included the location of the survey point, a morphological categorization of the specimen based on broad organism types (forb, tree, shrub, invertebrate, etc.), and an image. Notably, PP

was excluded from this campaign of collections due to time constraints. Based on satellite imagery, polygons of LR1-LR5 were constructed to enable a point in polygon analysis of our surveyed points using the UTM zone 16N projection. From here we were able to construct a summary dataset that contained area, organism counts, and biodiversity metrics for each revetment (Appendix A). To calculate our Shannon index, evenness, and Bray-Curtis mean (through Bray-Curtis dissimilarity), we used these equations:

Shannon Index:

$$H' = - \sum (p_i \times \log p_i)$$

Evenness:

$$E = \frac{H'}{\log \alpha}$$

Bray-Curtis Dissimilarity:

$$BC_{ij} = \frac{\sum |x_{ik} - x_{jk}|}{\sum (x_{ik} + x_{jk})}$$

Bray-Curtis Mean:

$$\text{Mean } BC_i = \frac{1}{n-1} \sum_{j \neq i} BC_{ij}$$

2.4 *Community Reconstruction*

When assembling the constructed ecosystems, four factors were considered: community composition on a general level, community composition on a species level, predation relationships between species, and similarities with Chicago ecosystems characterized in literature. The community composition on a general level was available for each LR from our ArcGIS survey. The community composition on a species level was drawn from our DNA and morphological analysis of the specimens found in each site. Literature sources were used to determine predation relationships between identified species. The sources consulted for predation relationship were the Cary Institute's Who Eats What guide, Bianca Kreuzinger-Janik and colleagues 2022 paper on predation in moss ecosystems, the Indiana Department of Environmental Management's fact sheet on algae ecosystems, and the Ellis et al. 2021 review of lichen ecology (Morganstern et al., 1994; Svensson and Rydin, 2022; Indiana Department of Environmental Management, n.d. Ellis et al., 2021). There were also multiple local sources which were used to compare field observations with ecosystems charac-

terized in the literature. These sources were the Illinois Government’s review of Chicago’s River and Lake Shore, the Chicago Lake Plain Ecoregion database, and the Chicago Wilderness Atlas of Biodiversity (Illinois Department of Natural Resources, 2008; bplant.org, n.d.; Sullivan, 1997). By utilizing sources on local ecosystems and predation relationships, species which were not specifically identified in the field can be associated with the revetment based on their relationships with the known species.

3 Results and Analysis

3.1 Genetic Biodiversity Measures

We investigated genetic biodiversity in three ways: by comparing the taxa present across revetments and looking at phylogenetic diversity, by analyzing mismatch counts between our sequences and DNA Subway voucher sequences, and by examining the native or exotic status of the sampled organisms.

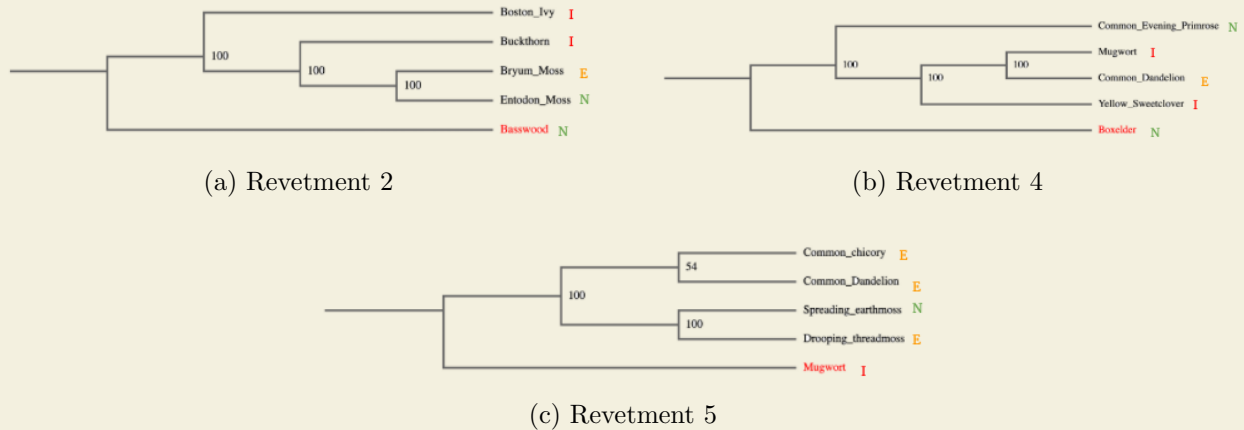


Figure 2: Phylogenetic trees by revetment

Phylogenies of plants from LR2 (Fig. 2a), LR4 (Fig. 2b), and LR5 (Fig. 2c) showed a broad range of taxa per revetment. (We lacked a sufficient number of high quality plant samples from LR1 and LR3 to construct phylogenies for them.) Revetment 2 harbored species from five plant orders, revetment 4 from four, and revetment 5 from three, indicating substantial phylogenetic diversity at each site. To test whether different revetment types filter species by evolutionary lineage, we analyzed the phylogeny of all successfully sequenced forbs (Fig. 3). Samples from

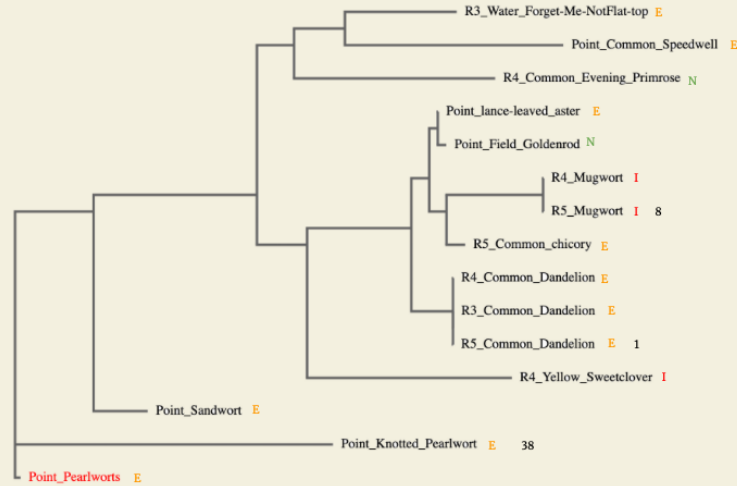


Figure 3: Forbs across LR and PP

different revetments were broadly distributed across the tree, with no clustering by revetment type, suggesting a lack of phylogenetic filtering across the different environments. We analyzed the phylogeny of all mosses and the phylogeny of all invertebrates similarity and again found no phylogenetic signal by revetment type (Appendix B).

We also examined genetic biodiversity using mismatch counts between our sequences and reference voucher sequences in DNA Subway. Most of our samples had zero mismatches (23 out of 35), ten had <10 mismatches, and two samples, Fissidens moss (*Fissidens neomagofukui*) and Knotted pearlwort (*Sagina nodosa*) had <20 (21 and 38, respectively) (Fig. 3). Especially notable were the instances of organisms sampled from multiple different revetments because this allowed us to analyze intraspecific variation between revetments. Mugwort (*Artemisia vulgaris*) samples from LR4 and LR5 differed significantly: the sequence from LR5 had 8 mismatches while the sequence from revetment 4 had none. Similarly, the common dandelion (*Taraxacum officinale*) sample from LR5 had 1 mismatch, whereas those from LR3 and LR4 had zero. The Asian ladybird beetle sampled from PP had no mismatches, but the one from LR5 had 1. These differences indicate the presence of intraspecific genetic diversity within species across revetments, even with limited sampling.

Interestingly, in each of these cases the organisms with the non-zero mismatches occurred in samples from LR5, which could suggest that this revetment hosts individuals with greater genetic divergence

from the reference sequences. While our sample size is much too limited to draw conclusions, this pattern could possibly reflect localized genetic variation or the presence of distinct source populations at LR5 and could be an interesting subject for further study.

An analysis of the native (N) or exotic (E) status of species samples sampled shows a strong difference by type: most moss species were native, and none of the exotic ones were listed as regionally invasive (I), whereas forbs and invertebrates were heavily dominated by exotic and invasive species. This pattern may be important to consider when evaluating which revetment types are most best for supporting biodiversity. Given that mosses appear to harbor a higher proportion of native species compared to forbs or invertebrates, revetments that promote moss diversity may be especially valuable for conserving the region's native biodiversity and limiting the dominance of exotic species.

3.2 Constructed Ecosystems

The constructed ecosystems allow for qualitative comparisons between the ecologies of the different revetments, as each ecosystem is different.

The PP constructed ecosystem is a moss-based ecosystem with a significant amount of pollinator resources further back from the shoreline. At the upper trophic levels, there are both primary and secondary consumers, all of which are invertebrates. In moss-based ecosystems, however, there is often an imbalance between predators and prey, similar to soil-based ecosystems (Svensson and Rydin, 2022).

There are also invasive species present, some of which do not play a noticeable role in the ecosystem. The revetment with the most similar ecosystem to PP was LR5.

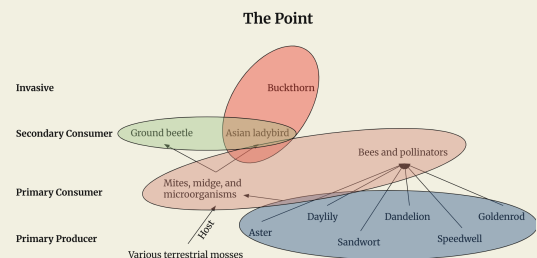


Figure 4: PP constructed ecosystem.

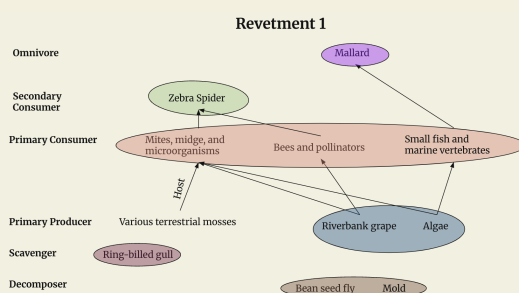


Figure 5: LR1 constructed ecosystem.

LR1's constructed ecosystem was the most marine of all the revetments, with intertidal algae and coastal birds contributing to the ecosystem. The presence of scavengers is similar to LR2; however, the lack of

surface colonizers on the large area of exposed rock created a lot of similarities with LR4. Out of all the revetments, LR1 promoted the most avian activity.

LR2 was unique in that it contained significant amounts of lichen, canopy providers, and invertebrates. The jagged terrain of LR2 provided ample room for invertebrates, while the rocks provided the surface for lichen and moss to grow. The semi-unconnected nature of this ecosystem left room for invasive species, mostly canopy providers.

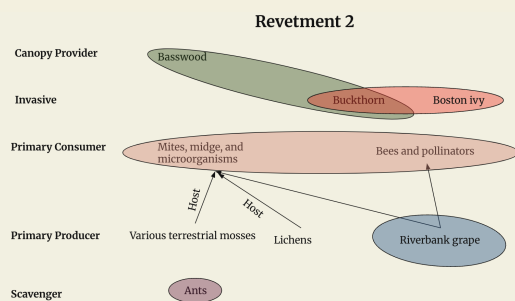
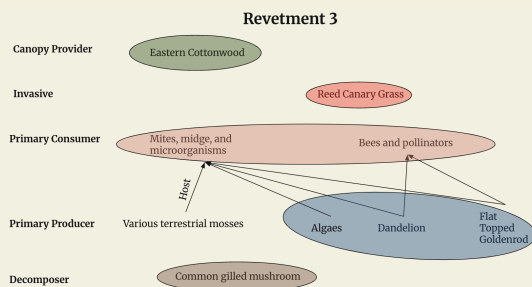
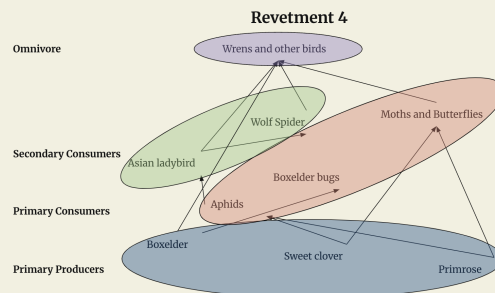


Figure 6: LR2 constructed ecosystem.

Compared to the other revetments, LR3 contains a primarily soil-based ecosystem, with mushrooms present as decomposers and a significant amount of pollinator resources. While algae is still present, it plays a smaller role than in other ecosystems. Just as in LR2, gaps in the ecosystem and community provide space for canopy providers and invasive surface colonizers. LR2 and LR3 have largely similar ecosystems, however there are more mosses and lichens present in LR2.



(a) LR3 constructed ecosystem



(b) LR4 constructed ecosystem

LR4 is the only ecosystem that does not contain mosses, and the transition between pebble beach and prairie creates a foredune ecosystem (Sullivan, 1997). The trophic levels are clear and well-defined in this ecosystem, with small birds serving as omnivores to keep insect populations in control. LR4 has similarities with LR1 in level of vegetation, however its ecosystem diagram is quite different.

LR5 contains a very similar ecosystem to PP. Both LR5 and PP rely on mosses as surface colonizers and invertebrates as secondary consumers. The main difference between the two ecosystems is that the pollinator resources are different, as LR5 contains chicory and mugwort, both of which are only found in this revetment. LR5 is thus not entirely a moss-based ecosystem, as there are many forbs present.

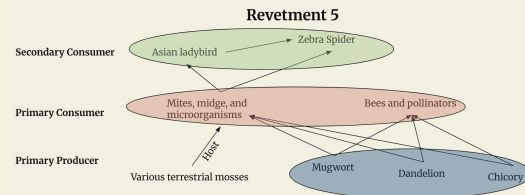
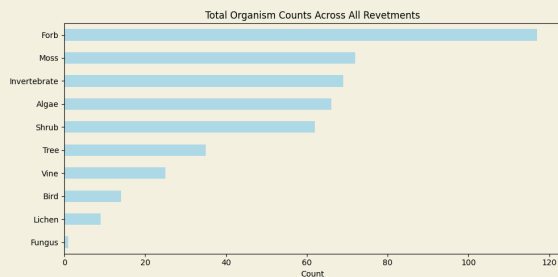


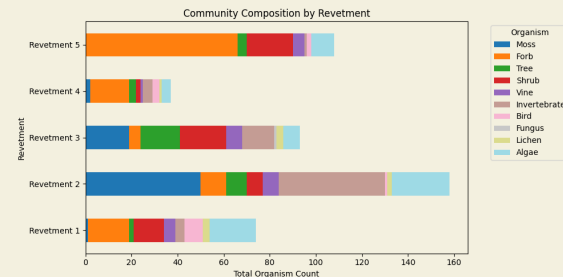
Figure 8: LR5 constructed ecosystem.

3.3 Organism Category Biodiversity Metrics

To assess biodiversity across revetments, we analyzed organismal group composition at each site. Rather than identifying individual species, we classified samples into broad categories—including mosses, algae, shrubs, and invertebrates—and quantified diversity using richness, richness density, Shannon index diversity, and Bray-Curtis dissimilarity. We also visualized patterns of similarity and difference using choropleth maps, non-metric multidimensional scaling (NMDS), and radial plots.



(a) Total organism category counts across all revetments.



(b) Organism category composition per revetment.

Figure 9: Bar chart visualizations of organism abundance and composition.

Organism counts show that forbs—defined here as herbaceous, flowering, broadleaf plants—were the most dominant category, followed by mosses and invertebrates (Fig. 9a). LR5 was especially forb-dominated, likely due to the abundance of soil-trapping crevices between its cubic concrete blocks. Invertebrates were most prevalent in LR2, where large angular rocks created ample gaps for spiderwebs and landing surfaces. LR2 also promoted moss growth, further supported by its higher moss abundance shown in Fig. 9b. These composition patterns hint at possible ecological

filtering and spatial structuring driven by revetment material and design—trends that will be further visualized using Bray-Curtis dissimilarity.

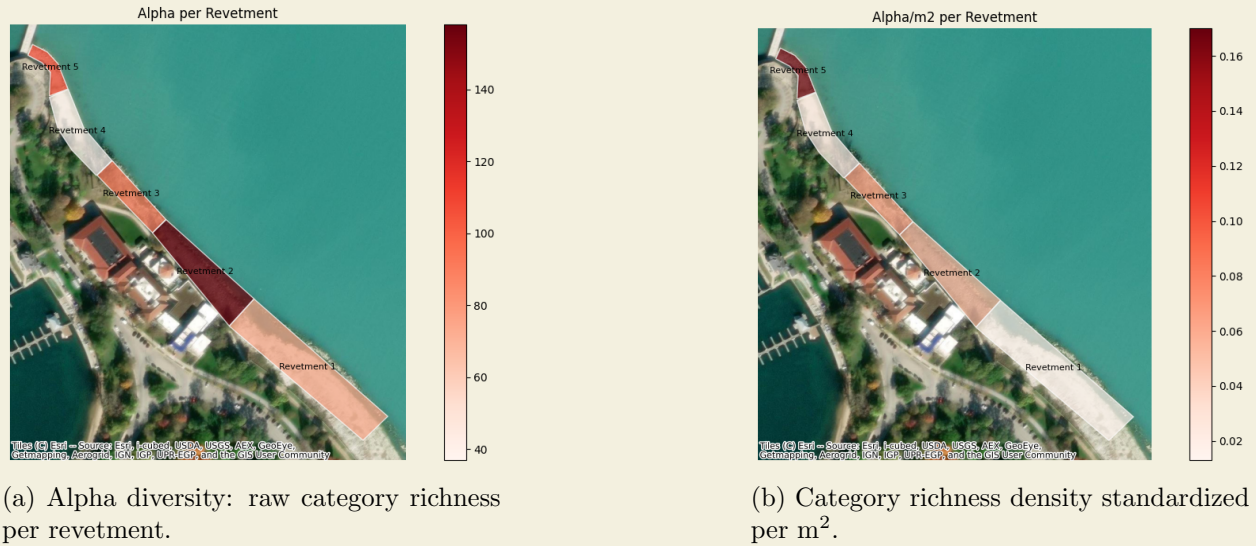


Figure 10: Alpha diversity metrics across revetments.

Raw alpha diversity—defined as the number of distinct organism categories—ranged from low richness at LR4 to high richness at LR2. When standardized per square meter, LR5 showed the highest richness density by far, with all other revetments converging toward similar densities (Fig. 10a).

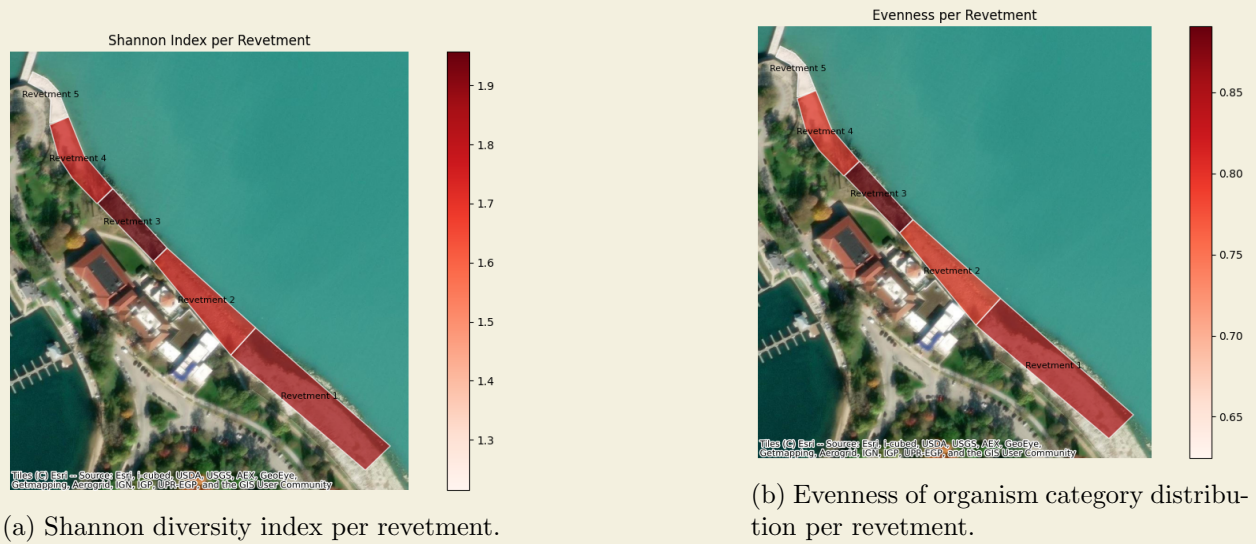


Figure 11: Shannon diversity and evenness of community composition.

Shannon diversity and evenness further explore species richness by incorporating dominance. LR5 stands out as an ecosystem dominated by a single category, showing high richness density but low evenness. In contrast, LR3 appears to be the most balanced ecosystem with a relatively high Shannon diversity index. LR1, despite having low richness density, surprisingly exhibits high evenness (Fig. 11a).

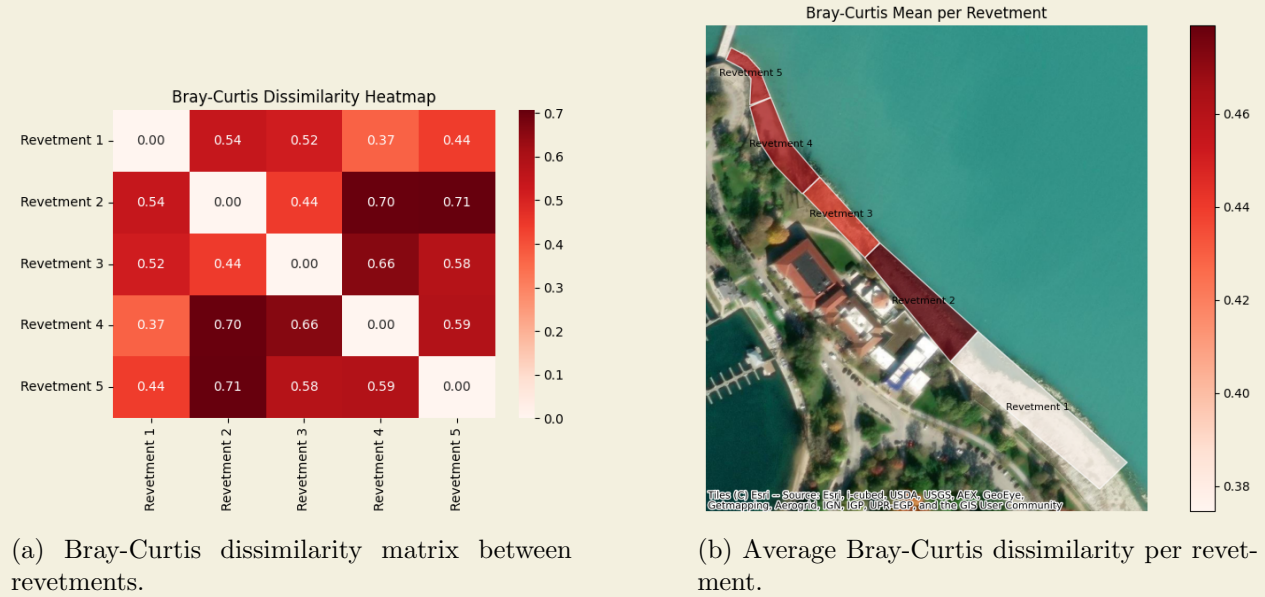


Figure 12: Community dissimilarity between revetments using Bray-Curtis metrics.

To quantify how distinct revetments were in community composition, we used Bray-Curtis dissimilarity. LR5 and LR2 were the most dissimilar, while LR1 and LR4 were the most similar (Fig. 12a). The choropleth map of mean dissimilarity (Fig. 12b) highlights LR2 as the most distinct revetment overall, reinforcing the influence of revetment geometry on shaping ecological communities.

The NMDS plot (Fig. 13) further visualizes these dissimilarities. LR2 and LR3 cluster close together, as do LR1 and LR4, while LR5 stands apart with its distinct community composition.

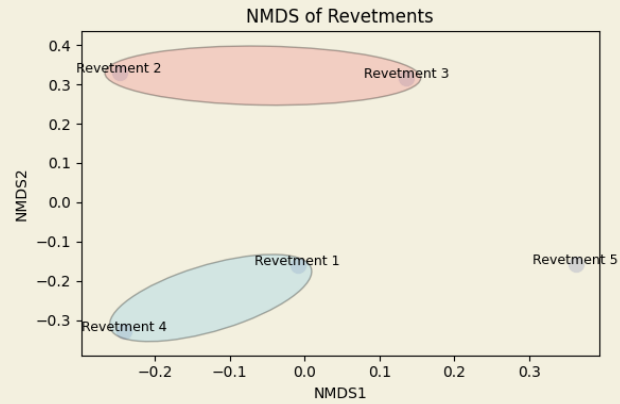


Figure 13: NMDS plot of Bray-Curtis dissimilarity values between revetments.



Figure 14: Radial biodiversity “fingerprints” for each revetment, summarizing richness, density, Shannon index, evenness, and Bray-Curtis distinctness.

Finally, radial plots for each revetment (Fig. 14) visualize overall biodiversity profiles. Similar radial shapes—like those of LR2 and LR3 or LR1 and LR4—suggest comparable ecological structures. LR5 again stands out for its highly uneven and specialized composition. These visual and quantitative analyses collectively reveal how revetment design significantly shapes community richness and structure—not just in terms of what groups are present, but how they coexist and differentiate across space.

4 Limitations

Before arriving at the implications tied to our results, there must be a disclaimer on the limitations that factored into possible sources of inaccuracy or skewed results on this project.

In regards to our sampling methods, there were certainly inconsistencies we could not control for. For example, time of day, temperature, humidity, cloud cover, and wind speeds were all variables that were not controlled for when sampling. This may have an impact on the accuracy of our results, as invertebrates are mobile and have preference to weather conditions, possibly taking shelter during non-ideal conditions to specific species. Additionally, all observations were done by eye in a constrained time setting, leading to inaccuracies from human error due to line of sight missing, or being rushed to return to the classroom.

One certain limitation for our project was the lack of time to collect a morphological survey from PP. This limited our ability to compare species richness, Shannon Index, evenness, Bray-Curtis statistics and more between PP and LR sites.

In the lab, we also had limitations in identifying certain species genetically, with some samples yielding inconclusive sequences. The rates of inconclusive sequences could have been increased by the time constraints to process all of the samples in the three hour class periods. In addition, morphological identification was challenging for many of the samples as the method of storing the moss and lichen in tubes produced generic green slush when stored in the fridge.

5 Conclusion and Implications

The synthesis of our approaches to data interpretation have allowed for a deeper understanding of the complexity of biodiversity in the area. The question of revetments impacting species density and ecosystem types has become more clear at this point.

The most important numerical value we were able to arrive at was the Shannon Index for each individual revetment. With these values, we were able to not only discern the species richness of LR1-LR5, but also the evenness of different species type across these revetments. Coupling this with the Bray-Curtis Dissimilarity statistic provided with a baseline analysis of which revetment site harbored the most species richness in relevance to evenness, while also comparing the differences

between the individual revetment sites. Based on these results specifically, LR2 and LR3 were most effective at harboring diverse and relatively dense biodiversity, while also being the second most insular pair of sites besides for LR1 and LR4. LR3 did harbor a higher Shannon Index value, and a lower Bray-Curtis Dissimilarity than LR2.

One of the broader aims of this project was to see if it was possible to determine if one specific revetment type could be considered objectively better than the rest. With methods built in class, it was possible to constrain one factor very well, that being which revetment types best fostered biodiversity. While this was going on, one member of our group took further note of observations of the sites being studied, based on knowledge from work done specifically on coastal erosion.

These observations were used to estimate the general efficiency of each site from LR1-LR5 at combating erosion of the shoreline the revetments were backed by. The ideal revetments to disperse tidal energy were found to be LR2 and LR3, both of which took the form of rubble based revetments, with angular blocks being arranged almost randomly, with spaces in between. The difference in materials between revetments two and three are noticeable in appearance, but the effect on durability is unknown. However, it is clear that in comparison to the other revetment sites at LR that these two were most efficient at preventing erosion.

Looking at cost of implementation, it is extremely difficult to pin down any numbers in specificity, however it can be assumed that the cheapest of the sites that was implemented was LR3, as it likely was just the arrangement of discarded concrete.

Based on these three assumptions, it is very possible to constrain what would be the best revetment in general to implement to maximize the benefits of the factors discussed above. LR3 is likely most cost effective, second most efficient at preventing erosion, and the site with the highest Shannon Index value. Following LR3 is LR2, with the best effective erosion prevention, the second best Shannon Index, but an unknown general estimate of implementation. In addition to this there is the possibility that this concrete is reused material, providing a secondary use to ripped up rubble which is another benefit of using LR3 as a revetment in the future.

The constructed ecosystem diagrams are useful for making qualitative comparisons between the revetments; however, they shouldn't be used as normative. In different situations, different ecosys-

tems may be more beneficial for the greater community. The main takeaway should be that revetments like PP and LR2 create moss-based ecosystems while shorelines like LR4 create room for more forb-centered ecologies. Using the ecosystem diagrams should be used as one part of the conversation, along with the biodiversity and erosion measures.

5.1 Next Steps for Further Study

If given the opportunity to expand on this project given more time and tools, there are a few steps that would be ideal to move towards initially. First off, it would be ideal to go through multiple surveys of each site, at different times of day, under different conditions, across different climate conditions. With this, we could normalize the values collected for all quantitative analysis that was based off of our ArcGIS survey.

Secondly, it would be beneficial to go back to these sites and collect samples that were more difficult to see or obtain. By this, I refer to small invertebrates like insects and arachnids, as well as avian species, and all other invertebrates which may fear human interactions, like raccoons or mice as examples. To do this, we could leave transect viewing cameras out for 24 hours, and see what larger species we can discern passing through the line of sight at 0, 10, and 30 foot distances.

In the lab, it would be beneficial to go back and retest the samples we found to be inconclusive. This would allow for more extensive species identification and deeper genetic diversity analysis. This would require significantly more resources however, so improved morphological analysis could also lead to improved results.

To build on the genetic biodiversity aspect of this project, future work could include sequencing multiple individuals per species to better assess intraspecific genetic variation. A larger sample size of conspecific samples would allow us to make stronger conclusions about the genetic biodiversity between revetments and look for potential differences between them. In addition, incorporation of functional traits and environmental variables could help clarify the ecological composition of each revetment.

Another step to take would be to find exact or numerical measurements for cost effectiveness, durability, material type, shape, and efficiency at preventing erosion. To do this, more research

into the history of construction, sifting through paperwork for material and cost would be effective. For shape, simply measuring a random selection of rubble, and averaging the volume would be a good estimate. Durability would best be tested using something like a small form wave tank left on, measuring force hitting each materials and measuring disintegration of material by mass change, To measure coastal erosion, a more long term project would be necessary, taking repeated aerial and at scene images of the sites, to see changes in land area.

Many of these steps are not overly complicated, and given more time, could provide much more dimensionality to our project, allowing us to inch closer to finding a likely answer to our broadest question of is there a specific revetment type that is objectively the best.

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A Revetment Results and Metrics from Survey 123 Form

Table 1: Biodiversity and Community Metrics by Revetment

LR	Area (m ²)	Moss	Forb	Tree	Shrub	Vine	Invert.	Bird	Fungus	Lichen	Algae	Alpha
LR 1	5676.21	1	18	2	13	5	4	8	0	3	35	89
LR 2	2993.49	50	11	9	7	7	46	1	0	2	25	158
LR 3	1433.59	19	5	17	20	7	14	0	1	3	7	93
LR 4	1818.07	2	17	3	2	1	4	3	0	1	4	37
LR 5	635.38	0	66	4	20	5	1	2	0	0	10	108

LR	Alpha/m ²	Shannon Index	Evenness	Bray-Curtis Mean	Material
LR 1	0.0157	1.739	0.791	0.403	Silurian Dolostone
LR 2	0.0528	1.727	0.786	0.476	Large mafic igneous angular rock
LR 3	0.0649	1.957	0.891	0.449	Large concrete angular rubble
LR 4	0.0204	1.756	0.799	0.479	Pebble/Cobble Beach
LR 5	0.1700	1.215	0.624	0.471	Large cubic concrete blocks

B Other Phylogenies

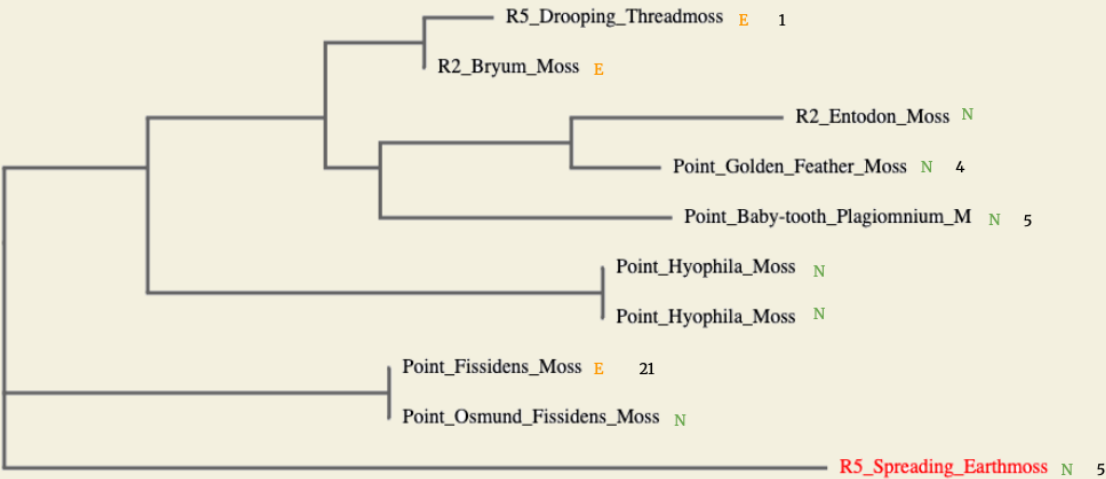


Figure 15: Mosses across LR and PP

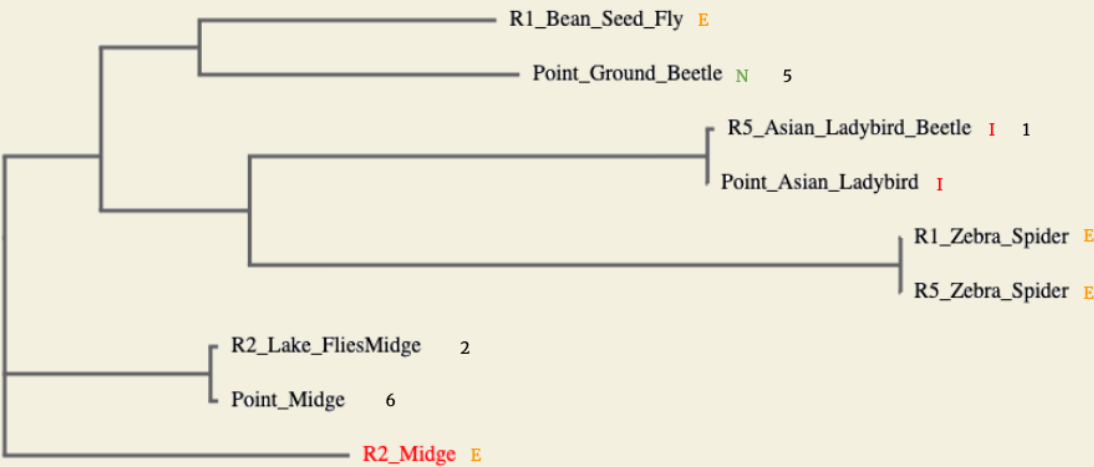


Figure 16: Invertebrates across LR and PP

C Pictures of Each Revetment



(a) LR 1



(b) LR 2



(c) LR 3



(d) LR 4



(e) LR 5

Figure 17: Photographs at each revetment site (LR1–LR5).